Ansible – Roles.

A **role** in Ansible is a structured way to organize playbooks, variables, files, templates, and tasks into a reusable unit.

Why Roles Exist

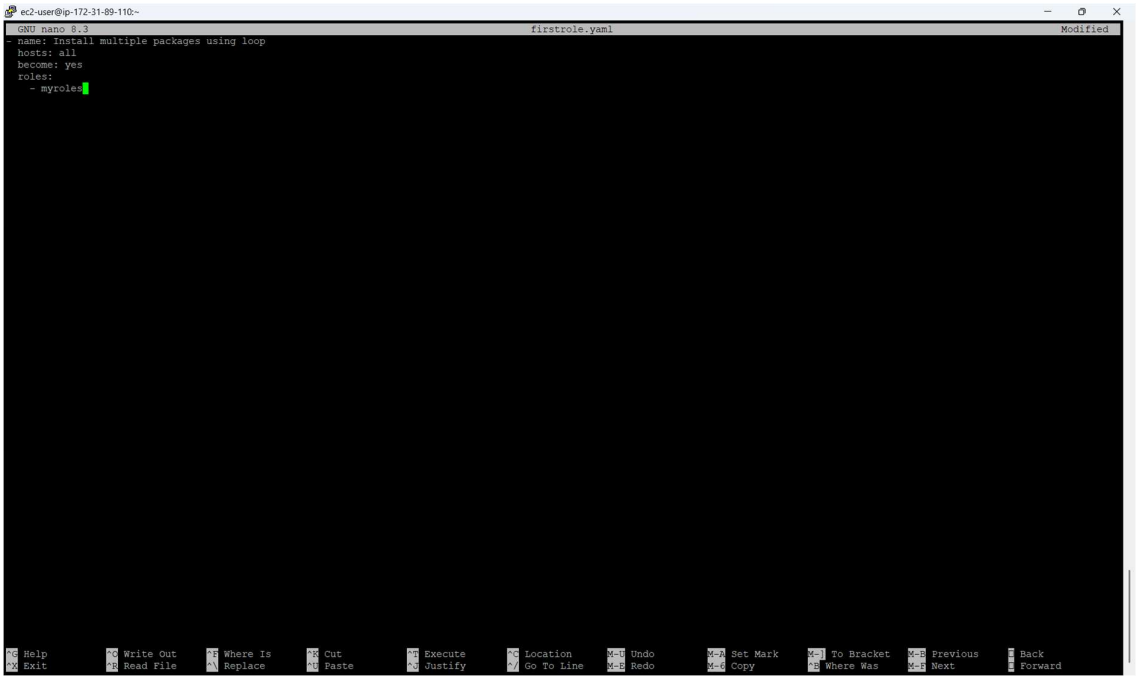
Without roles:

- One huge YAML file
- Repeated code
- Hard to maintain
- Impossible to scale

With roles:

- Modular structure
- Reusable across projects
- Clean separation of concerns
- Easier debugging

This is the example file for creating users in the slave machine.



This is the command to create an role.

```
[ec2-user@ip-172-31-91-97 ~]$ ansible-galaxy init myrole
- Role myrole was created successfully
```

The default directory of the role.

```
0 directories, 0 files
[ec2-user@ip-172-31-89-110 firstrole]$ ls
README.md  defaults  files  handlers  meta  tasks  templates  tests  vars
[ec2-user@ip-172-31-89-110 firstrole]$ tree

├── README.md
├── defaults
│   └── main.yml
├── files
├── handlers
│   └── main.yml
├── meta
│   └── main.yml
├── tasks
│   └── main.yml
├── templates
├── tests
│   ├── inventory
│   └── test.yml
└── vars
    └── main.yml

8 directories, 8 files
[ec2-user@ip-172-31-89-110 firstrole]$
```

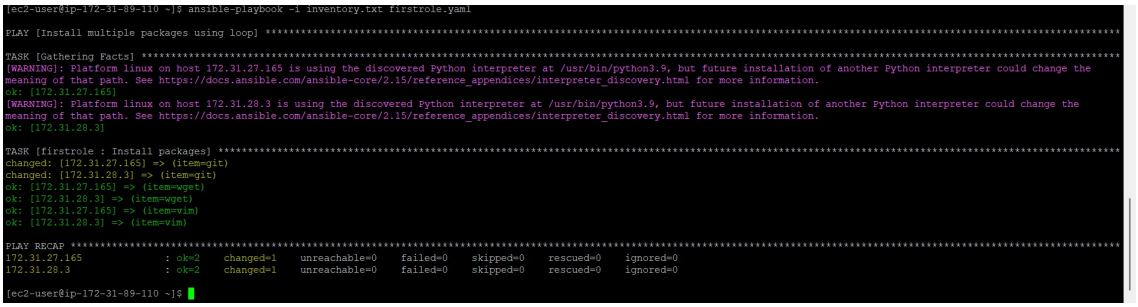
We have to create an role file in the tasks to put the tasks in the directory.

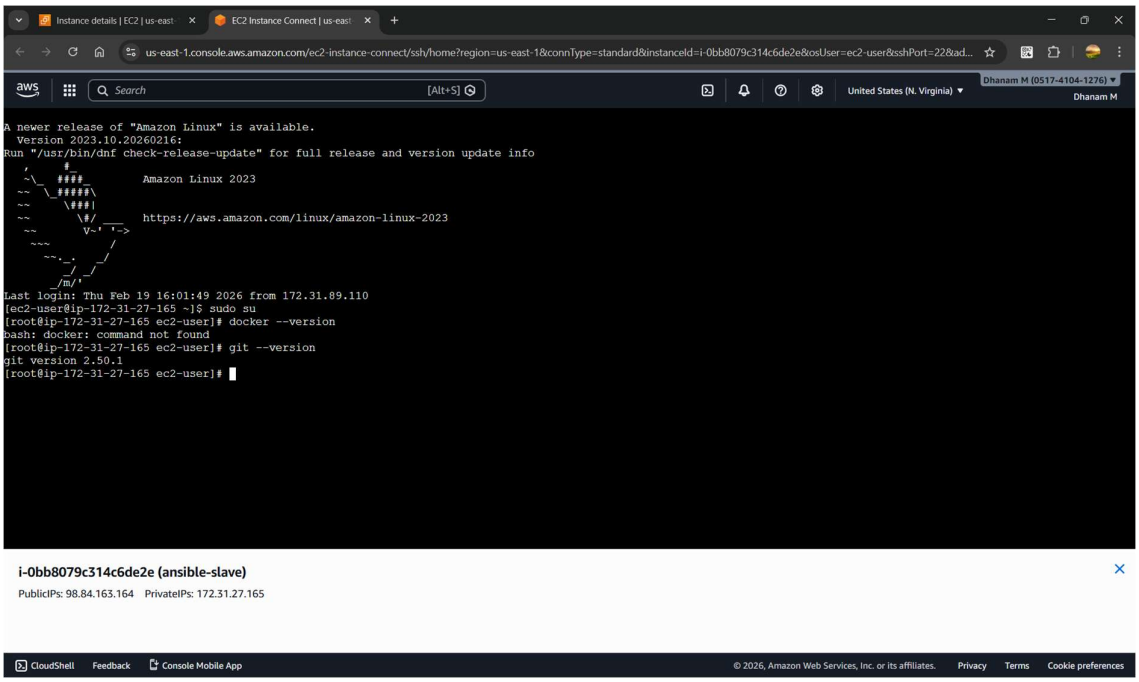
```
[ec2-user@ip-172-31-91-97 ~]$ cd myrole/
[ec2-user@ip-172-31-91-97 myrole]$ cd tas
-bash: cd: tas: No such file or directory
[ec2-user@ip-172-31-91-97 myrole]$ cd tasks/
[ec2-user@ip-172-31-91-97 tasks]$ nano main.yml
```

```
ec2-user@ip-172-31-89-110:~/firstrole/tasks
  GNU nano 8.3                                   main.yml
---
# tasks file for firstrole
  - name: Install packages
    yum:
      name: "{{ item }}"
      state: present
    loop:
      - git
      - wget
      - vim

                                   [ Read 10 lines ]
^G Help      ^O Write Out   ^W Where Is    ^K Cut       ^T Execute   ^C Location   M-U Undo     M-A Set Mark   M-] To Bracket  M-B Previous   ^_ Back
^X Exit      ^R Read File   ^\ Replace     ^U Paste     ^J Justify   ^/ Go To Line  M-E Redo     M-6 Copy       ^B Where Was    M-F Next       ^_ Forward
```

Then we have to use the ansible command to call the file.





The process have been executed successfully.